HWA CHONG INSTITUTION
JC2 Preliminary Examination
Higher 1

CANDIDATE NAME		**CT GROUP**	
CENTRE NUMBER		**INDEX NUMBER**	

GEOGRAPHY

8834/01

Paper 1

11 September 2024

3 hours

Additional Materials: Insert

READ THESE INSTRUCTIONS FIRST

If you have been given an Answer Booklet, follow the instructions on the front cover of the Booklet.
Write your Centre number, index number and name on the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer Question 1 in **Section A**.
Answer both questions in **Section B**.
Answer **two** questions in **Section C**.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

Section A

Cluster 2 Fieldwork

1 Four 17-year old students undertook an investigation about the perception of consumers of community responses to climate change in Singapore. They examined the perception of consumers by investigating the hypothesis -

'Consumers in Singapore see a stronger role for community response than for government and businesses in tackling climate change.'

Community response was defined as 'any action carried out by non-governmental and non-corporate stakeholders'.

The students collected primary data by conducting a survey in May and did so by asking their friends to participate in an online survey and sought their help to forward the link to family and friends. When they were unable to reach their targeted sample size of 100 respondents, the students decided to expand their number of responses for the survey by conducting the survey individually outside a busy shopping mall - Bugis Junction - on a Wednesday and Thursday afternoon. Passers-by were approached to do the survey.

In addition to the survey, the students conducted face-to-face interviews with 6 family members and neighbours, including two elderly neighbours. The students also conducted some research on consumers' perception on climate change and found some data from a perception survey conducted by the National Climate Change Secretariat (NCCS) in 2019.

Resource 1 shows street-level views outside Bugis Junction. Resource 2 shows surveyed Singapore consumers' perception of different actors' importance in fighting climate change. Resource 3 shows 2019 selected survey findings by NCCS about Singapore consumers' perception on the responsibility of different actors in fighting climate change.

(a) Explain why the hypothesis is appropriate for the investigation. [3]

(b) With reference to Resource 1, suggest the challenges that the students may face conducting the survey online and at Bugis Junction. [4]

(c) With reference to Resource 1, suggest how the students could minimise potential risks when administering the survey at Bugis Junction. [5]

(d) Explain the ethical considerations that the students should take note of when conducting the interviews with family members and neighbours. [4]

(e) Compare the consumers' perception of the actors' importance in fighting climate change shown in Resource 2. [4]

(f) The students concluded that consumers in Singapore think that community response to climate change is more important than the government and businesses. With reference to Resources 1, 2 and 3, evaluate the validity of their conclusion. [10]

[Turn over

Section B

Cluster 1 Sustainable Future and Climate Change

Answer **both** questions

2 Resource 4 shows urban growth in selected cities. Resource 5 shows urban population annual growth rates in primary cities and suburbs in New York and Boston metropolitan areas 2010 – 2020. Resource 6 shows some details of the Deonar dumping ground fire in Mumbai, India and its location within the Mumbai Metropolitan Region. Resource 7 shows solid waste production in Mumbai and Sweden and solid waste management by Tekniska Verken in Sweden.

(a) Cite data to compare urban growth in selected cities shown in Resource 4. [3]

(b) With reference to Resource 4, explain how demand may be placed on natural environments in Lagos and Mumbai. [4]

(c) With reference to Resource 5, suggest the possible challenges faced by New York and Boston in progressing towards sustainable urban development. [4]

(d) With reference to Resource 6, explain the limitations of landfills as a waste management strategy. [6]

(e) With reference to Resource 7, suggest how less demand is placed on the environment when waste is viewed as a potential resource. [5]

3 Resource 8 shows changes in global surface temperature between 1880 to 2020. Resource 9 shows effects of major drivers on biodiversity in Earth's ecosystems. Resource 10 shows global warming and the tipping points of Earth's climate systems.

(a) Cite data from Resource 8 to describe the changes in global surface temperature for annual average and natural drivers only between 1880 and 2020. [4]

(b) With reference to Resource 9, compare the effect of habitat change and climate change on biodiversity in ecosystems. [5]

(c) Explain how ecosystems may be affected by contemporary climate change. [4]

(d) With reference to Resource 10, describe the main effects of global warming on the Earth's climate systems. [4]

(e) With reference to Resources 8, 9 and 10, suggest how changes to aquatic and terrestrial ecosystems can impact humans. [5]

[Turn over

Section C

Answer **two** questions

4 'Successful urban reimaging strategies must include consideration of sustainable urban development'.

Evaluate this statement. [13]

5 'Given the rate of contemporary climate change, mitigation measures are more urgent than adaptation measures'.

Evaluate this statement. [13]

6 Evaluate the influence of human activities on climate change. [13]

Acknowledgements
Question 1 Resource 1 *@ https://www.capitaland.com/sg/*
Question 1 Resource 1 *@ https://www.todayonline.com/singapore/*
Question 1 Resource 2 *@ https://www.mdpi.com/2071-1050/14/19/12330*
Question 1 Resource 3 *@ https://www.nccs.gov.sg/*
Question 2 Resource 4 *@ https://www.internetgeography.net/topics/what-is-urbanisation/*
Question 2 Resource 5 *@ https://www.brookings.edu/*
Question 2 Resource 6 *@ https://www.ibtimes.co.in/photos/*
Question 2 Resource 7 *@ https://asiaconverge.com/2016/03/669/*
Question 3 Resource 8 *@ https://www.ipcc.ch/report/ar6/wg1/figures/summary-for-policymakers*
Question 3 Resource 9 *@ https://www.semanticscholar.org/paper/*
Question 3 Resource 10 *@ https://www.pik-potsdam.de/en/output/infodesk/tipping-elements*